

Resonance Energy Systems

Tagline: `*(TBD — see `branding/taglines/TAGLINES_MASTER.md`)*`

Document: HB-01 — Turbine Selection Doctrine (Crossflow-first)

Version: 0.1

Date: 2026-01-22

Applicable contexts: Creek / Canal / Outfall (default)

See ``04_Technology/handbooks/_shared/DEFAULT_CONTEXTS.md``

Disclaimer & Positioning

- This handbook is engineering doctrine for deployment and operations.
- Performance claims must be supported by measured head, flow, losses, and efficiency.
- Visionary inspirations may influence design heuristics but do not replace validated physics.

Turbine Selection Doctrine (Crossflow-first)

Handbook ID: HB-01

Version: 0.1 (Draft)

Deployment Contexts (Default: All Three)

This handbook is written to be valid for three default deployment environments:

- **[CREEK]** Creek / foothills run-of-river (cold + debris)
- **[CANAL]** Irrigation canal drop (high flow + ops crew)
- **[OUTFALL]** Industrial outfall / water-works (24/7 + compliance + corrosion)

Context overlays live in: ``contexts/CONTEXT_OVERLAYS.md``.

Shared definitions: ``04_Technology/handbooks/_shared/DEFAULT_CONTEXTS.md``.

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1) Overview

(Draft)

2) Key Principles

(Draft)

3) Deployment Contexts (Default: All Three)

(This section will be inserted automatically if missing.)

4) Procedures

(Draft)

5) Failure Modes & Red Flags

(Draft)

6) Checklists

See `checklists/`.

7) References

See `11\_References/MASTER\_REFERENCES.md`.